

Matrices & Vectors



Addition and scalar multiplication

- 1 Let A be the matrix with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and B be the matrix with rows $\langle -1, 4 \rangle$ and $\langle 0, 2 \rangle$. Find $A + B$ and $3A$.
 - 2 Using the same A and B , find $A - 2B$.
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Matrix multiplication

- 3 Using A (rows $\langle 2, 1 \rangle, \langle 3, -1 \rangle$) and B (rows $\langle -1, 4 \rangle, \langle 0, 2 \rangle$) from above, find AB .
 - 4 Let A have rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and let $\mathbf{v} = \langle 2, -3 \rangle$. Find $A\mathbf{v}$.
 - 5 Explain, without computing BA , why AB and BA need not be equal for 2×2 matrices.
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Determinant and inverse

- 6 Find $\det A$ for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and state whether A is invertible.
 - 7 Find A^{-1} for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$.
 - 8 Without computing it, explain why the matrix with rows $\langle 4, 2 \rangle$ and $\langle 6, 3 \rangle$ has no inverse.
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Solving systems with the inverse matrix

- 9 Solve the system $3x - y = 8$ and $x + 2y = 5$ using $\mathbf{x} = M^{-1}\mathbf{b}$.
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Vector magnitude and dot product

- 10 For $\mathbf{u} = \langle 3, 4 \rangle$ and $\mathbf{w} = \langle -4, 3 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.
 - 11 What does $\mathbf{u} \cdot \mathbf{w} = 0$ tell you about $\mathbf{u}$ and $\mathbf{w}$ from the previous question?
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Angle between vectors

- 12 Find the angle between $\mathbf{p} = \langle 1, 0 \rangle$ and $\mathbf{q} = \langle 1, 1 \rangle$.
- 13 Find the angle between $\langle 6, -8 \rangle$ and $\langle 3, 4 \rangle$, rounded to the nearest tenth of a degree.
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Multiple choice — easy

- 14 Find $\det A$ for the matrix A with rows $\langle -2, 9 \rangle$ and $\langle -5, -1 \rangle$.
- a. 19
 - b. 47
 - c. -43
 - d. 49
- 15 Find $\det A$ for the matrix A with rows $\langle 4, 8 \rangle$ and $\langle -4, 6 \rangle$.
- a. -64
 - b. 58
 - c. 56
 - d. -8
- 16 Find $\det A$ for the matrix A with rows $\langle 2, -2 \rangle$ and $\langle 9, -8 \rangle$.
- a. 5
 - b. -34
 - c. 2
 - d. 4

- 17 Find $\det A$ for the matrix A with rows $\langle 0, -4 \rangle$ and $\langle 1, 4 \rangle$.
- a. -4
 - b. 4
 - c. 16
 - d. 6
- 18 Find $\det A$ for the matrix A with rows $\langle -4, -8 \rangle$ and $\langle -1, -3 \rangle$.
- a. 20
 - b. -20
 - c. 4
 - d. 6
- 19 Find $\det A$ for the matrix A with rows $\langle -1, -4 \rangle$ and $\langle 5, -1 \rangle$.
- a. -19
 - b. 23
 - c. -9
 - d. 21
- 20 Find $\det A$ for the matrix A with rows $\langle 0, -8 \rangle$ and $\langle -4, 4 \rangle$.
- a. -30
 - b. 32
 - c. -29
 - d. -32

- 21 Find $\det A$ for the matrix A with rows $\langle -5, 3 \rangle$ and $\langle -8, 6 \rangle$.
- a. -4
 - b. -54
 - c. 22
 - d. -6
- 22 Find $\det A$ for the matrix A with rows $\langle -5, 0 \rangle$ and $\langle -7, 8 \rangle$.
- a. -40
 - b. 35
 - c. -37
 - d. -38
- 23 Find $\det A$ for the matrix A with rows $\langle -1, -3 \rangle$ and $\langle -9, 9 \rangle$.
- a. -34
 - b. -36
 - c. 36
 - d. 18
- 24 Find $\det A$ for the matrix A with rows $\langle 7, -5 \rangle$ and $\langle 6, -4 \rangle$.
- a. 2
 - b. 4
 - c. -58
 - d. 22

- 25 Find $\det A$ for the matrix A with rows $\langle -2, 9 \rangle$ and $\langle -7, -1 \rangle$.
- a. 67
 - b. -61
 - c. 65
 - d. 23
- 26 Find $\det A$ for the matrix A with rows $\langle -7, 9 \rangle$ and $\langle 6, -7 \rangle$.
- a. -5
 - b. 103
 - c. -3
 - d. 21
- 27 Find $\det A$ for the matrix A with rows $\langle -1, -5 \rangle$ and $\langle 2, -3 \rangle$.
- a. -7
 - b. 15
 - c. -17
 - d. 13
- 28 Find $\det A$ for the matrix A with rows $\langle 5, 2 \rangle$ and $\langle -9, -3 \rangle$.
- a. -33
 - b. 3
 - c. 5
 - d. -39

Multiple choice — medium

- 29 Let A have rows $\langle 4, 2 \rangle$ and $\langle 3, 7 \rangle$, and let $\mathbf{v} = \langle 5, 2 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 26, 29 \rangle$
 - b. $\langle 24, 27 \rangle$
 - c. $\langle 29, 24 \rangle$
 - d. $\langle 24, 29 \rangle$
- 30 Let A have rows $\langle -3, 7 \rangle$ and $\langle 1, 3 \rangle$, and let $\mathbf{v} = \langle 3, -5 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -42, -12 \rangle$
 - b. $\langle -12, -44 \rangle$
 - c. $\langle -44, -12 \rangle$
 - d. $\langle -44, -14 \rangle$
- 31 Let A have rows $\langle -4, 4 \rangle$ and $\langle 8, 4 \rangle$, and let $\mathbf{v} = \langle 5, 3 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -6, 52 \rangle$
 - b. $\langle 52, -8 \rangle$
 - c. $\langle -8, 50 \rangle$
 - d. $\langle -8, 52 \rangle$
- 32 Let A have rows $\langle 1, 1 \rangle$ and $\langle 7, -7 \rangle$, and let $\mathbf{v} = \langle 7, 7 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 14, 0 \rangle$
 - b. $\langle 0, 14 \rangle$
 - c. $\langle 14, -2 \rangle$
 - d. $\langle 16, 0 \rangle$

- 33 Let A have rows $\langle 9, -6 \rangle$ and $\langle -4, 3 \rangle$, and let $\mathbf{v} = \langle -2, -5 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 12, -9 \rangle$
 - b. $\langle -7, 12 \rangle$
 - c. $\langle 14, -7 \rangle$
 - d. $\langle 12, -7 \rangle$
- 34 Let A have rows $\langle -1, 6 \rangle$ and $\langle 2, 2 \rangle$, and let $\mathbf{v} = \langle -1, 5 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 33, 8 \rangle$
 - b. $\langle 31, 8 \rangle$
 - c. $\langle 8, 31 \rangle$
 - d. $\langle 31, 6 \rangle$
- 35 Let A have rows $\langle 9, -9 \rangle$ and $\langle -1, 7 \rangle$, and let $\mathbf{v} = \langle -8, -8 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 2, -48 \rangle$
 - b. $\langle 0, -48 \rangle$
 - c. $\langle 0, -50 \rangle$
 - d. $\langle -48, 0 \rangle$
- 36 Let A have rows $\langle -7, -6 \rangle$ and $\langle 8, 4 \rangle$, and let $\mathbf{v} = \langle -5, -4 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 61, -56 \rangle$
 - b. $\langle 59, -56 \rangle$
 - c. $\langle 59, -58 \rangle$
 - d. $\langle -56, 59 \rangle$

- 37 Let A have rows $\langle -9, -8 \rangle$ and $\langle 2, -4 \rangle$, and let $\mathbf{v} = \langle 9, -1 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -73, 22 \rangle$
 - b. $\langle 22, -73 \rangle$
 - c. $\langle -71, 22 \rangle$
 - d. $\langle -73, 20 \rangle$
- 38 Let A have rows $\langle -2, -6 \rangle$ and $\langle 7, -5 \rangle$, and let $\mathbf{v} = \langle -9, -3 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 38, -48 \rangle$
 - b. $\langle 36, -48 \rangle$
 - c. $\langle 36, -50 \rangle$
 - d. $\langle -48, 36 \rangle$
- 39 Let A have rows $\langle -8, -1 \rangle$ and $\langle -7, -9 \rangle$, and let $\mathbf{v} = \langle 9, -9 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -63, 16 \rangle$
 - b. $\langle -61, 18 \rangle$
 - c. $\langle 18, -63 \rangle$
 - d. $\langle -63, 18 \rangle$
- 40 Let A have rows $\langle -8, -4 \rangle$ and $\langle 3, 6 \rangle$, and let $\mathbf{v} = \langle -4, -5 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -42, 52 \rangle$
 - b. $\langle 52, -44 \rangle$
 - c. $\langle 54, -42 \rangle$
 - d. $\langle 52, -42 \rangle$

- 41 Let A have rows $\langle -7, -2 \rangle$ and $\langle 1, 0 \rangle$, and let $\mathbf{v} = \langle 8, -1 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -54, 6 \rangle$
 - b. $\langle 8, -54 \rangle$
 - c. $\langle -52, 8 \rangle$
 - d. $\langle -54, 8 \rangle$
- 42 Let A have rows $\langle -6, -4 \rangle$ and $\langle -5, -2 \rangle$, and let $\mathbf{v} = \langle 7, 9 \rangle$. Find $A\mathbf{v}$.
- a. $\langle -76, -53 \rangle$
 - b. $\langle -78, -53 \rangle$
 - c. $\langle -78, -55 \rangle$
 - d. $\langle -53, -78 \rangle$
- 43 Let A have rows $\langle 3, -2 \rangle$ and $\langle -5, 1 \rangle$, and let $\mathbf{v} = \langle 5, 5 \rangle$. Find $A\mathbf{v}$.
- a. $\langle 5, -22 \rangle$
 - b. $\langle 7, -20 \rangle$
 - c. $\langle 5, -20 \rangle$
 - d. $\langle -20, 5 \rangle$

Multiple choice — hard

- 44 Find A^{-1} for the matrix A with rows $\langle 4, -5 \rangle$ and $\langle -5, 6 \rangle$.
- a. $\langle -6, -5 \rangle, \langle -5, -4 \rangle$
 - b. $\langle 4, -5 \rangle, \langle -5, 6 \rangle$
 - c. $\langle -5, -5 \rangle, \langle -5, -4 \rangle$
 - d. $\langle -5, -4 \rangle, \langle -6, -5 \rangle$

45 Find A^{-1} for the matrix A with rows $\langle 3, 1 \rangle$ and $\langle -1, 0 \rangle$.

a. $\langle 0, -1 \rangle, \langle 1, 3 \rangle$

b. $\langle 0, 1 \rangle, \langle -1, 3 \rangle$

c. $\langle 1, 3 \rangle, \langle 0, -1 \rangle$

d. $\langle 3, 1 \rangle, \langle -1, 0 \rangle$

46 Find A^{-1} for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle 3, 1 \rangle$.

a. $\langle -1, 3 \rangle, \langle 1, -2 \rangle$

b. $\langle 2, 1 \rangle, \langle 3, 1 \rangle$

c. $\langle -1, 1 \rangle, \langle 3, -2 \rangle$

d. $\langle 3, -2 \rangle, \langle -1, 1 \rangle$

47 Find A^{-1} for the matrix A with rows $\langle 1, -2 \rangle$ and $\langle 2, -5 \rangle$.

a. $\langle 5, 2 \rangle, \langle -2, -1 \rangle$

b. $\langle 2, -1 \rangle, \langle 5, -2 \rangle$

c. $\langle 5, -2 \rangle, \langle 2, -1 \rangle$

d. $\langle 1, -2 \rangle, \langle 2, -5 \rangle$

48 Find A^{-1} for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle -5, -3 \rangle$.

a. $\langle 3, -5 \rangle, \langle 1, -2 \rangle$

b. $\langle 2, 1 \rangle, \langle -5, -3 \rangle$

c. $\langle 3, 1 \rangle, \langle -5, -2 \rangle$

d. $\langle -5, -2 \rangle, \langle 3, 1 \rangle$

- 49 Solve the system $6y = -30$ and $6x = 24$ using $\mathbf{x} = M^{-1}\mathbf{b}$.
- a. $\langle -5, 4 \rangle$
 - b. $\langle 6, -5 \rangle$
 - c. $\langle 4, -5 \rangle$
 - d. $\langle 4, -7 \rangle$
- 50 Solve the system $4x - 5y = -3$ and $-6x + 4y = -6$ using $\mathbf{x} = M^{-1}\mathbf{b}$.
- a. $\langle 3, 1 \rangle$
 - b. $\langle 3, 3 \rangle$
 - c. $\langle 6, 3 \rangle$
 - d. $\langle 5, 3 \rangle$
- 51 Solve the system $x + 3y = -15$ and $x + 2y = -8$ using $\mathbf{x} = M^{-1}\mathbf{b}$.
- a. $\langle 6, -7 \rangle$
 - b. $\langle -7, 6 \rangle$
 - c. $\langle 8, -7 \rangle$
 - d. $\langle 6, -9 \rangle$
- 52 Solve the system $2x - y = 5$ and $6x + 6y = -48$ using $\mathbf{x} = M^{-1}\mathbf{b}$.
- a. $\langle -1, -7 \rangle$
 - b. $\langle 1, -7 \rangle$
 - c. $\langle -1, -9 \rangle$
 - d. $\langle -7, -1 \rangle$

53 Solve the system $x - y = 0$ and $-3x - 6y = -54$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle 9, 6 \rangle$

b. $\langle 6, 4 \rangle$

c. $\langle 6, 6 \rangle$

d. $\langle 8, 6 \rangle$

54 For $\mathbf{u} = \langle 20, 21 \rangle$ and $\mathbf{v} = \langle -5, -12 \rangle$, find $\cos \theta$ where θ is the angle between them, using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{352}{377}$

b. $-\frac{352}{377}$

c. $-\frac{27}{29}$

d. $-\frac{176}{21}$

55 For $\mathbf{u} = \langle -20, -21 \rangle$ and $\mathbf{v} = \langle 8, -15 \rangle$, find $\cos \theta$ where θ is the angle between them, using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{155}{493}$

b. $-\frac{155}{493}$

c. $\frac{156}{493}$

d. $\frac{155}{46}$

56 For $\mathbf{u} = \langle 20, -21 \rangle$ and $\mathbf{v} = \langle -5, -12 \rangle$, find $\cos \theta$ where θ is the angle between them, using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{152}{377}$

b. $\frac{76}{21}$

c. $\frac{153}{377}$

d. $-\frac{152}{377}$

57 For $\mathbf{u} = \langle 8, 15 \rangle$ and $\mathbf{v} = \langle 7, 24 \rangle$, find $\cos \theta$ where θ is the angle between them, using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{417}{425}$

b. $\frac{208}{21}$

c. $-\frac{416}{425}$

d. $\frac{416}{425}$

58 For $\mathbf{u} = \langle 7, -24 \rangle$ and $\mathbf{v} = \langle 5, 12 \rangle$, find $\cos \theta$ where θ is the angle between them, using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{253}{325}$

b. $-\frac{253}{38}$

c. $-\frac{252}{325}$

d. $-\frac{253}{325}$